

Clarification on Oracle-OR Algorithm

Let $\{\mathcal{P}_r\}_{r=1}^R$ be the post-NMS prediction sets from the R single-resolution models, and let

$$\mathcal{G} = \{(b_g, c_g)\}$$

denote the ground-truth set.

For each ground-truth instance $g = (b_g, c_g)$, we define the candidate set:

$$\mathcal{M}_g = \left\{ p = (b_p, c_p, s_p) \in \bigcup_{r=1}^R \mathcal{P}_r : \text{IoU}(b_p, b_g) \geq \tau \right\}.$$

We then define the correctly classified subset:

$$\mathcal{M}_g^{\text{cls}} = \{p \in \mathcal{M}_g : c_p = c_g\}.$$

If $\mathcal{M}_g^{\text{cls}} \neq \emptyset$, we retain this subset; otherwise, we keep $\mathcal{M}_g$. The selected prediction is:

$$p^*(g) = \arg \max_{p \in \mathcal{M}_g} \text{IoU}(b_p, b_g).$$

Predictions that do not intersect any ground-truth instance are treated as false alarms, with an additional redundancy filtering step applied to avoid counting overlapping false detections multiple times.

Interpretation: recall and classification upper bounds

Under this definition, the oracle provides:

- A **recall upper bound**: a ground-truth instance is detected if *any* single-resolution model detects it.
- A **classification upper bound (conditional on detection)**: when evaluating classification accuracy (e.g., confusion matrices), only matched predictions are considered. Since the oracle retains a correctly classified prediction whenever one exists in $\mathcal{M}_g$, it achieves the best possible classification performance given the detections.
- **No upper bound on mAP**: although the oracle maximizes recall by construction, it aggregates false positives from all resolutions. An additional IoU-based suppression step removes part of the redundant detections, but the total number of false positives increases, which degrades precision.

$$\text{Precision}_{\text{oracle}} = \frac{\text{TP}}{\text{TP} + \text{FP}_{\text{oracle}}}.$$

Since mAP depends on the full precision-recall curve and on the ranking induced by confidence scores (which remain unchanged), **the oracle is not guaranteed to maximize mAP**.

Input: Post-NMS prediction sets $\{\mathcal{P}_r\}_{r=1}^R$ from the R single-resolution models, where each prediction $p = (b_p, c_p, s_p)$ contains a bounding box b_p , a class label c_p , and a confidence score s_p ;
Ground-truth set $\mathcal{G} = \{(b_g, c_g)\}_{g=1}^{N_G}$;
IoU threshold τ .
Output: Oracle prediction set $\mathcal{P}_{\text{oracle}}$;
Oracle false-alarm set $\mathcal{F}_{\text{oracle}}$.

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 $\mathcal{P}_{\text{all}} \leftarrow \bigcup_{r=1}^R \mathcal{P}_r$ ;
 $\mathcal{P}_{\text{oracle}} \leftarrow \emptyset$ ;
 $\mathcal{P}_{\text{matched}} \leftarrow \emptyset$ ;
foreach  $(b_g, c_g) \in \mathcal{G}$  do
     $\mathcal{M}_g \leftarrow \emptyset$ ; // Predictions intersecting ground truth  $g$ 
    foreach  $(b_p, c_p, s_p) \in \mathcal{P}_{\text{all}}$  do
        if  $\text{IoU}(b_p, b_g) \geq \tau$  then
            add  $(b_p, c_p, s_p)$  to  $\mathcal{M}_g$ ;
        end
    end
    if  $\mathcal{M}_g \neq \emptyset$  then
         $\mathcal{M}_g^{\text{cls}} \leftarrow \{(b_p, c_p, s_p) \in \mathcal{M}_g : c_p = c_g\}$ ; // Class filtering
        if  $\mathcal{M}_g^{\text{cls}} \neq \emptyset$  then
             $\mathcal{M}_g \leftarrow \mathcal{M}_g^{\text{cls}}$ ;
        end
         $p^*(g) \leftarrow \arg \max_{(b_p, c_p, s_p) \in \mathcal{M}_g} \text{IoU}(b_p, b_g)$ ;
        add  $p^*(g)$  to  $\mathcal{P}_{\text{oracle}}$ ;
        add all elements of  $\mathcal{M}_g$  to  $\mathcal{P}_{\text{matched}}$ ;
    end
end
 $\mathcal{F}_{\text{raw}} \leftarrow \mathcal{P}_{\text{all}} \setminus \mathcal{P}_{\text{matched}}$ ; // Predictions with no intersection with any ground truth
 $\mathcal{F}_{\text{oracle}} \leftarrow \mathcal{F}_{\text{raw}}$ ;
foreach  $p_i = (b_i, c_i, s_i) \in \mathcal{F}_{\text{raw}}$  do
    foreach  $p_j = (b_j, c_j, s_j) \in \mathcal{F}_{\text{raw}}$  do
        if  $i \neq j$  and  $\text{IoU}(b_i, b_j) \geq \tau$  then
            if  $\text{area}(b_i) < \text{area}(b_j)$  then
                remove  $p_i$  from  $\mathcal{F}_{\text{oracle}}$ ;
            end
        end
    end
end
 $\mathcal{P}_{\text{oracle}} \leftarrow \mathcal{P}_{\text{oracle}} \cup \mathcal{F}_{\text{oracle}}$ ;
return  $\mathcal{P}_{\text{oracle}}, \mathcal{F}_{\text{oracle}}$ ;

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Algorithm 1: ORACLE-OR from post-NMS single-resolution predictions